Motherson Automotive & Technology engineering
Survery No 172/1-5,274/2,276/7,
Chengalpattu Sriperumbudur Road, Vadgal Village B Vallam Panchayat,,Vadagal,
Kanchipuram, Tamil Nadu, 602105

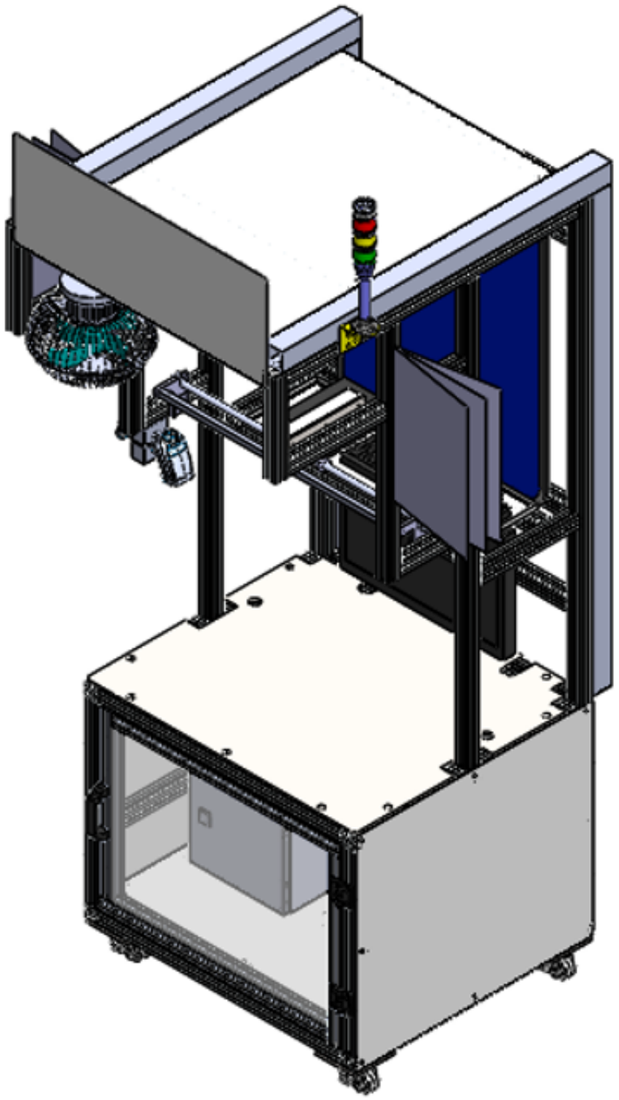
CUSTOMER NAME	MMLI PUNE
PROJECT NAME	S220 HL HV & LV FINAL INSPECTION STATION 11
PROJECT CODE	FMSPM00865
COMMISSION	ROBIS
PROJECT TYPE	WORK STATION
PLACE OF INSTALLATION	MMLI PUNE
RESPONSIBLE OF PROJECT	ROBIS
REVISION	02 (LIN Communication updated)
DESIGNER	MANORAJ.S

CREATED ON	26-07-2023	CHECKED BY		
LAST EDITED DATE	10-04-2024	CHECKED ON	29-07-2023	NUMBER OF PAGES 29

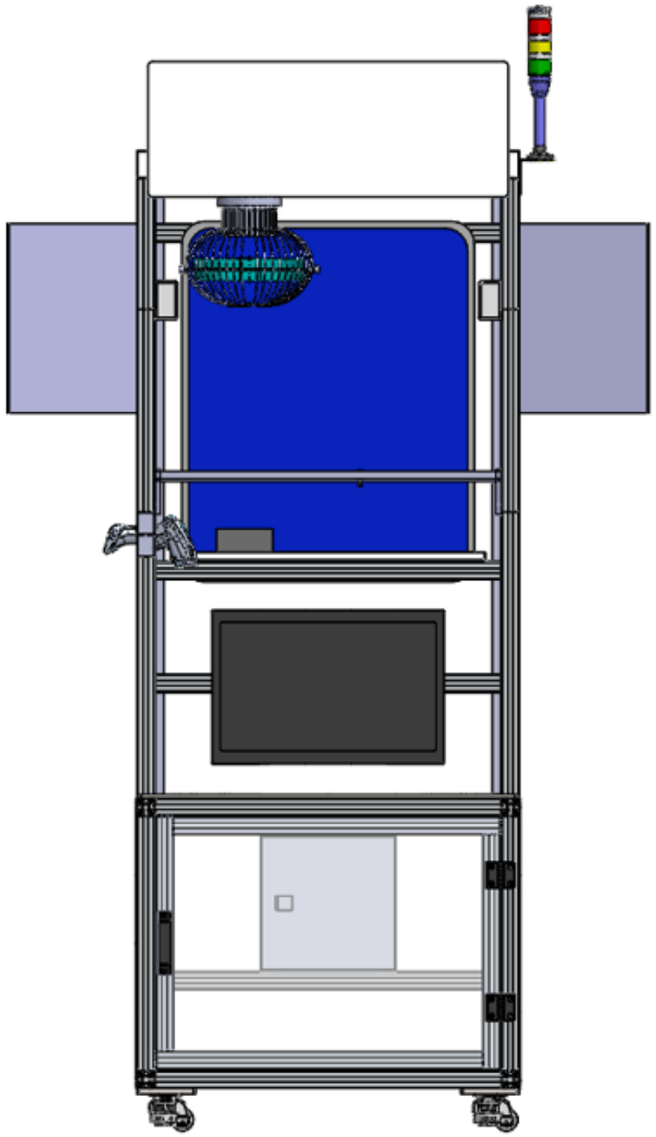
+CP/2

FMSPM00865 S220 HL _ High version + Low verion Final Inspection. station 11

motherson



ISO VIEW



FRONT VIEW

29

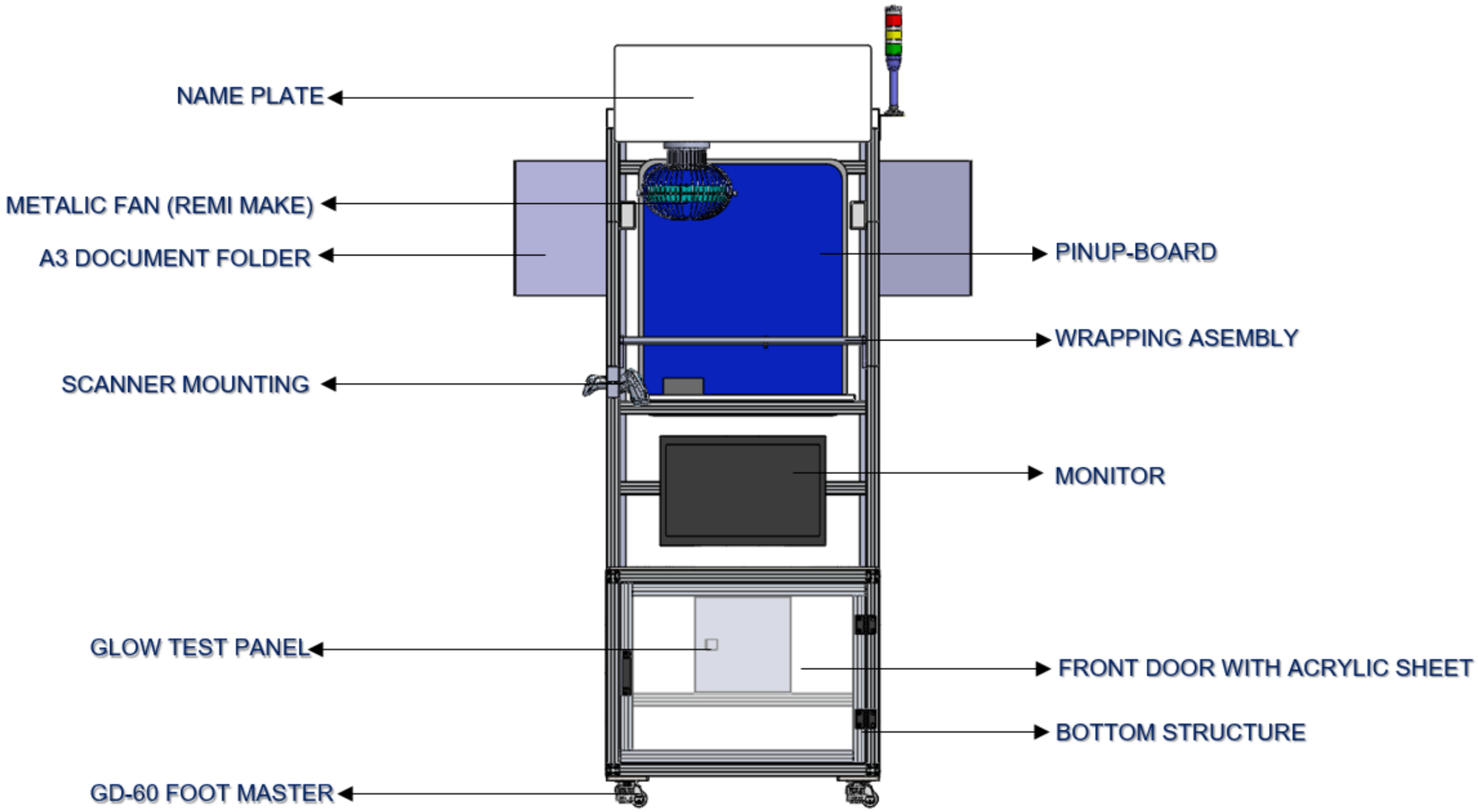
NOTE : PRINTER NOT CONSIDER

+/1

3

REVISION			Date	10-04-2024	ROBIS	Motherson Automotive & Technology engineering	MACHINE OVER VIEW		=	
			Ed.	MANORAJ.SELVARAJ					+ CP	
			Appr.		S220 HL HV & LV FINAL INSPECTION STATION 11				FMSPM00865	Page 2
Modification	Date	Name	Original		Replacement of	Replaced by				Page 2 / 29

FMSPM00865-COMPONENTS;

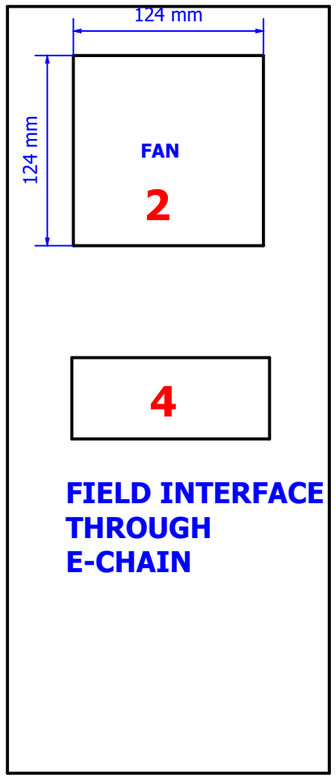


NOTE :GLOW TEST PANEL, PRINTER,TOWER LAMP NOT CONSIDER

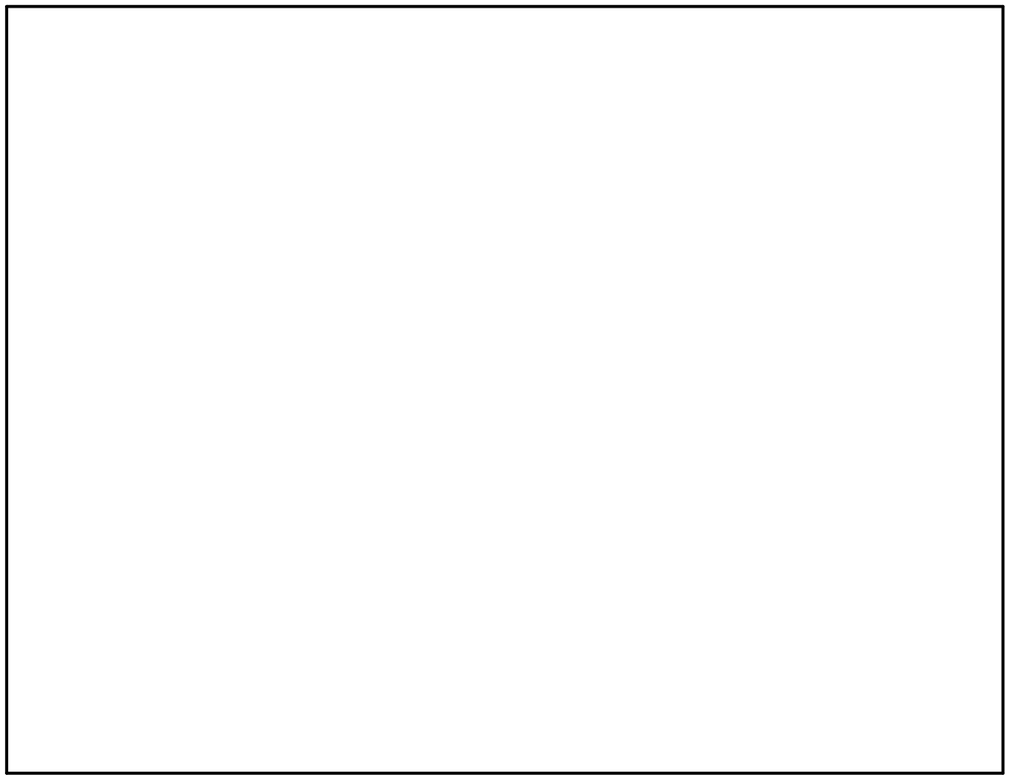
PANEL CUTOUT

CUSTOMIZED PANEL
H 500 X W 650 X D 210

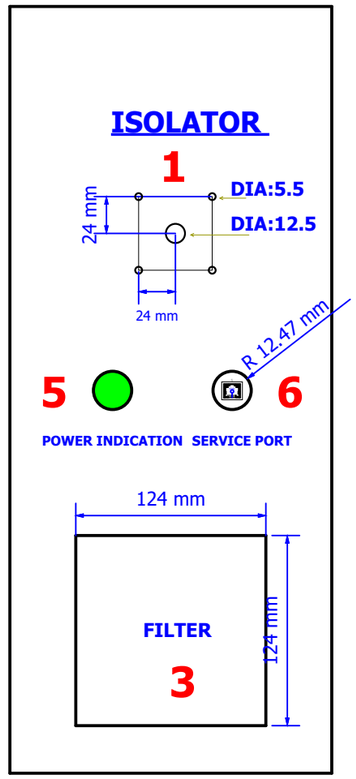
LEFT SIDE VIEW



FRONT SIDE VIEW



RIGHT SIDE VIEW



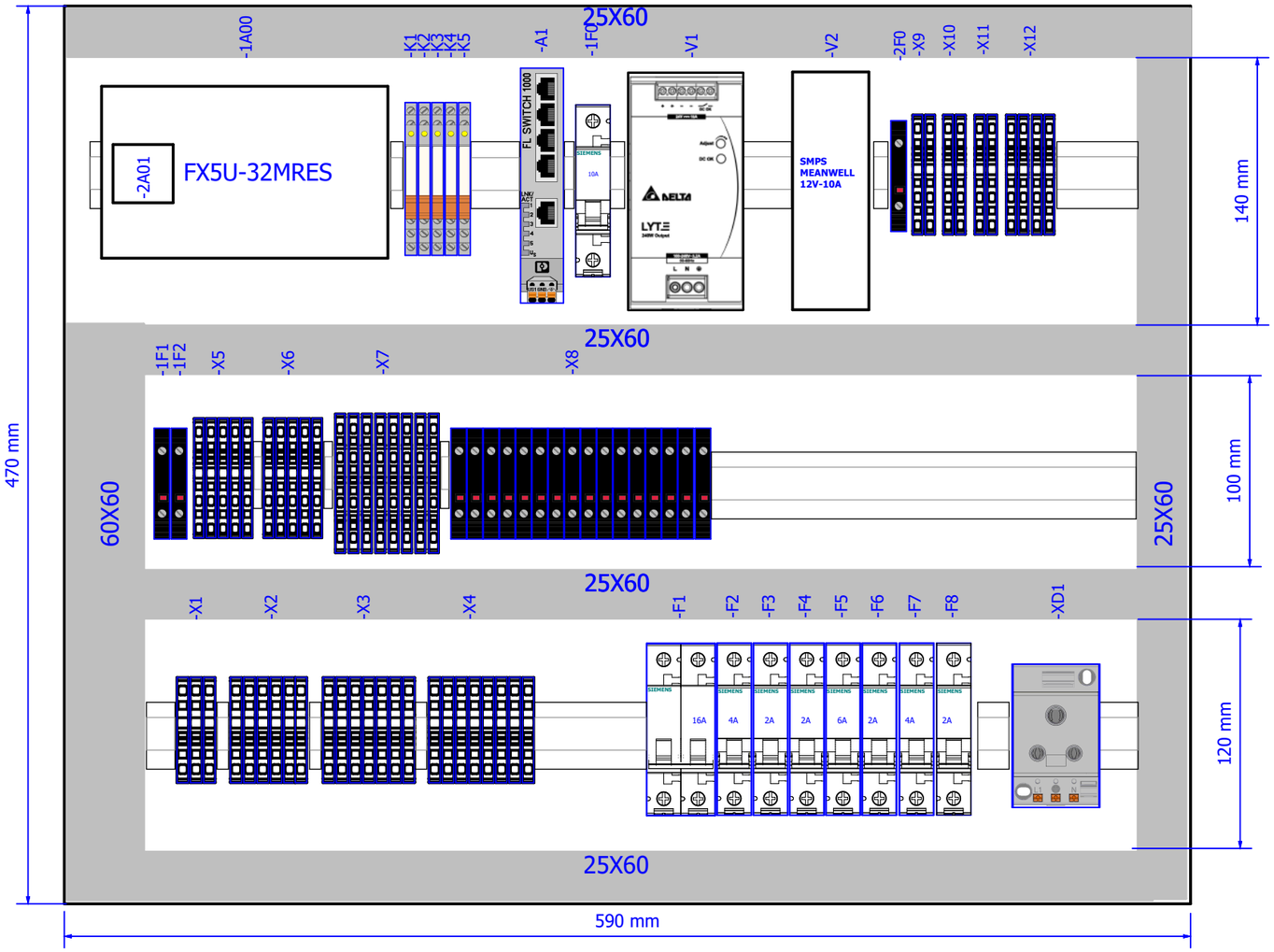
1	ISOLATOR CUTOUT
2	PANEL FAN CUTOUT
3	PANEL FILTER CUTOUT
4	FIELD INTERFACE CUTOUT
5	POWER INDICATION
6	SERVICE PORT

3 5

REVISION			Date	10-04-2024	ROBIS	Motherson Automotive & Technology engineering	PANEL CUTOUT		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV FINAL INSPECTION STATION 11			FMSPM00865	Page 4
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LAYOUT DIAGRAM

-1A00	CAAAZ00000000077 PLC FX5U32MRES MITSUBHISHI	1
-2A01	CAAAZ00000000076 RS COMMUNICATION CARD FX5-232-BD	1
-K1-K5	BMELES0935 SLIM RELAY COIL 24VDC	5
-A1	BMELES1310 ETEHRNET SWITCH 100 MB/S	1
-1F0	BMELES1708 MCB SINGLE POLE 10A SIEMENS MAKE	1
-V1	CAABA00000000488 PRO ECO 240W 24V 10A -WEIDMULLER SMPS	1
-1F1	BMELES0584 3075799 TB4-HESILED 24 (5X20) EI	1
-1F2	BMELES0584 3075799 TB4-HESILED 24 (5X20) EI	1
-X1	BMELES0589 TERMINAL BLOCK 4.0 SQ MM 1 IN 1 OUT	2
-X1	BMELES0595 TERMINAL BLOCK 4.0 SQ MM 1 IN 1 OUT YE/GN	1
-X2	BMELES0589 TERMINAL BLOCK 4.0 SQ MM 1 IN 1 OUT	6
-X3	BMELES0589 TERMINAL BLOCK 4.0 SQ MM 1 IN 1 OUT	7
-X4	BMELES0595 TERMINAL BLOCK 4.0 SQ MM 1 IN 1 OUT YE/GN	8
-X5	BMELES0364 TERMINAL BLOCK 2.5 SQ MM 1 IN 2 OUT	5
-X6	BMELES0364 TERMINAL BLOCK 2.5 SQ MM 1 IN 2 OUT	5
-X7	BMELES0366 TERMINAL BLOCK 2.5 SQ MM 2 IN 2 OUT	8
-X8	BMELES0584 3075799 TB4-HESILED 24 (5X20) EI	16
-X9	BMELES0364 TERMINAL BLOCK 2.5 SQ MM 1 IN 2 OUT	2
-X10	BMELES0364 TERMINAL BLOCK 2.5 SQ MM 1 IN 2 OUT	2
-X11	BMELES0364 TERMINAL BLOCK 2.5 SQ MM 1 IN 2 OUT	2
-X12	BMELES0364 TERMINAL BLOCK 2.5 SQ MM 1 IN 2 OUT	4
-F1	BMELES3072 SIEMENS MCB 2POLE 16AMPS 5SL62167RC	1
-F2	BMELES1706 MCB SINGLE POLE 4A SIEMENS MAKE	1
-F3	BMELES1705 MCB SINGLE POLE 2A SIEMENS MAKE	1
-F4	BMELES1705 MCB SINGLE POLE 2A SIEMENS MAKE	1
-F5	BMELES1707 MCB SINGLE POLE 6A SIEMENS MAKE	1
-F6	BMELES1705 MCB SINGLE POLE 2A SIEMENS MAKE	1
-F7	BMELES1706 MCB SINGLE POLE 4A SIEMENS MAKE	1
-F8	BMELES1705 MCB SINGLE POLE 2A SIEMENS MAKE	1
-2F0	BMELES0584 3075799 TB4-HESILED 24 (5X20) EI	1
XD1	CAABA00000000433 804000 INDIAN 6A DIN RAIL POWER SOCKET	1



REVISION			Date	10-04-2024	ROBIS	Motherson Automotive & Technology engineering	LAYOUT DIAGRAM		=
			Ed.	MANORAJ.SELVARAJ					+ CP
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PROJECT ARCHITECTURE



CAAAZ00000000077
PLC FX5U32MRES
MITSUBHISHI



CAAAZ00000000076
RS COMMUNICATION CARD
FX5-232-BD,MITSUBHISH



BMELES3189
Autonics
Panel Meter
MT4Y-DV-44



BMELES3035
Autonics
Panel Meter
MT4W-AA-41



BMELES1310
Ethernet Switch
100 MB/S



BMCONS0728
MOOTEK Industrial PC
MBOX-520-I5-M7-8265U



CAAAI00000000586
HMI MODEL : GS2107-WTBD
MITSUBHISH



BMELES3308
VIEW SONIC
22"Touch Monitor



BMELES3959
Data logic power
scanner PD9531

INPUT/OUTPUT

SAFETY

ETHERNET COMMUNICATION

RS 232 COMMUNICATION

RS 485 COMMUNICATION

5 7

REVISION			Date	10-04-2024	ROBIS	Motherson Automotive & Technology engineering	PROJECT ARCHITECTURE		=
			Ed.	MANORAJ.SELVARAJ					+ CP
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MAIN POWER DISTRIBUTION



-F1
BMELES3072
SIEMENS MCB 2POLE 16AMPS
5SL62167RC



-Q1
CAABA00000000274
16 A /3P /1WAY ISOLATOR
ROTARY SWITCH

L1 BK 2.5 mm² → -Q1 T1 → L1 OG 2.5 mm²
N1 BU 2.5 mm² → -Q1 T2 → N BU 2.5 mm²
-Q1 T3 → PE GNYE 2.5 mm²

-X1
L OG 2.5 mm²
N BU 2.5 mm²
PE GNYE 2.5 mm²

BMELES0589
TERMINAL BLOCK CX4
4.0 SQ MM 1 IN 1 OUT

-W0
3Gx2.5 mm²
BMELES1940
L BK/1 2.5 mm²
N BK/2 2.5 mm²
PE GNYE 2.5 mm²

BMELES3408
INDUSTRIAL
SOCKET LEGRAND
INDUSTRIAL SOCKET

POWER SUPPLY
1 PHASE 230V
50-60 Hz

-X2
1L1 BK 2.5 mm² → -Q1 T1 → L1 BK 2.5 mm²
1L1-7 BK 2.5 mm² → 1L1-7 / 14.1 -F6:2
1L1-8 BK/1 0.75 mm² → 1L1-8 / 23.7 -S2:13
1L1-10 BK 2.5 mm² → 1L1-10 / 8.9 -F7:2
1L1-11 BK 2.5 mm² → 1L1-11 / 9.1 -F8:2

BMELES0589
TERMINAL BLOCK CX4
4.0 SQ MM 1 IN 1 OUT

1L1-1 BK 2.5 mm² → -L1:x1 9.7 / 1L1-1
1L1-2 BK 2.5 mm² → -F3:2 8.3 / 1L1-2
1L1-3 BK 2.5 mm² → -F4:2 8.7 / 1L1-3
1L1-4 BK 2.5 mm² → -F5:2 9.5 / 1L1-4
1L1-5 BK/1 2.5 mm² → -S1-2F1:1 10.1 / 1L1-5
1L1-6 BK/1 2.5 mm² → -S2-2F3:1 10.6 / 1L1-6

-X3
1N1 BU 2.5 mm² → -Q1 T3 → N1 BU 2.5 mm²
1N1-8 BK/2 0.75 mm² → 1N1-8 / 23.5 -H3C:N
1N1-8 BK/2 0.75 mm² → 1N1-8 / 23.3 -H2C:N
1N1-9 BK/2 0.75 mm² → 1N1-9 / 23.8 -M2C:N
1N1-11 BK/2 1.5 mm² → 1N1-11 / 21.1 -VM1:7
1N1-12 BU 2.5 mm² → 1N1-12 / 9.2 -V2:N(-)

BMELES0589
TERMINAL BLOCK CX4
4.0 SQ MM 1 IN 1 OUT

1N1-1 BU 2.5 mm² → -L1:x2 9.8 / 1N1-1
1N1-2 BK/2 0.75 mm² → -M1 8.4 / 1N1-2
1N1-3 BU 1 mm² → -H1 8.6 / 1N1-3
1N1-4 BU 2.5 mm² → -XD1 9.5 / 1N1-4
1N1-5 BK/2 2.5 mm² → -S1:2 -S1:2 10.1 / 1N1-5
1N1-6 BK/2 2.5 mm² → -S2:2 10.6 / 1N1-6
1N1-7 BU 2.5 mm² → -1A00:N 14.1 / 1N1-7

-X4
PE GNYE 2.5 mm² → -Q1 T3 → PE GNYE 2.5 mm²
PE-8 GNYE 0.75 mm² → PE-8 / 24.2 -HMI:FG
PE-9 GNYE 1 mm² → PE-9 / 13.2 -A1:US:3
PE-10 GNYE 2.5 mm² → PE-10 / 9.2 -V2:PE

BMELES0595
TERMINAL BLOCK
4.0 SQ MM 1 IN 1 OUT YE/GN

PE-1 GNYE 2.5 mm² → -V1:- 8.1 / PE-1
PE-2 GNYE 0.75 mm² → -M1 8.4 / PE-2
PE-3 GNYE 2.5 mm² → -V2:- 9.2 / PE-3
PE-4 GNYE 2.5 mm² → -XD1 9.5 / PE-4
PE-5 GNYE 2.5 mm² → -S1:PE 10.2 / PE-5
PE-6 GNYE 2.5 mm² → -S2:PE 10.6 / PE-6
PE-7 GNYE 2.5 mm² → -1A00:PE 14.2 / PE-7
PE GNYE 2.5 mm² → PANEL EARTH

HMI PANEL EARTH
GNYE 2.5 mm²
PNEUMATIC PANEL EARTH
GNYE 2.5 mm²

6 8

REVISION			Date	10-04-2024	ROBIS	Motherson Automotive & Technology engineering	MAIN POWER DISTRIBUTION		=	
			Ed.	MANORAJ.SELVARAJ					+ CP	
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POWER SUPPLY

-W27.1
BMELES1938
3G1.5 sqmm cable

L01 / 11.3 -P1
N01 / 11.3 -P1
PE01 / 11.3 -P1

BK/1 1.5 mm²
BK/2 1.5 mm²
GNYE 1.5 mm²

L01
N01
PE01

-F2
MCB SINGLE POLE
4 A SIEMENS MAKE
5SL6104RC;
BMELES1706

2.5 mm² BK
L02

-V1
BMELES1532
DELTA LITE POWER SUPPLY
10A ,1 Phase ,240W

Weidmüller PROeco
L(+) N(-) PE
Adjust 22-28 V
+ + - -
13 14

1.5 mm² DBU 24VDC
1.5 mm² DBU P0VDC
2.5 mm² GNYE PE-1

-1F0:2 12.2 / 24VDC
-X6:1 12.6 / P0VDC
-X4:1:B 7.7 / PE-1

1L1-2 / 7.3 -X2:2:B
1N1-2 / 7.5 -X3:2:B
PE-2 / 7.7 -X4:2:B

2.5 mm² BK 1L1-2

-F3
BMELES1705
MCB SINGLE POLE
2A SIEMENS MAKE
5SL61027RC

BK/1 0.75 mm²
3L1-2

-W1
BMELES0075
3 G 0.75 mm²

0.75 mm² BK/1 2L1-2
0.75 mm² BK/2 1N1-2
0.75 mm² GNYE PE-2

-M1
BMELES0049
SK 3322107-
FAN AND FILTER UNIT

-LS
BMELES1669
AE PANEL DOOR
LIMIT SWITCH -
SZ 4127010

21
22

-W2
BMELES0075
3 G 0.75 mm²

0.75 mm² BK/2 4L1-2

-H1
BMELES2527
1Feet 6W LED
TUBE LIGHT WITH
ALUMINIUM BATTERY (CW)

1N1-3 / 7.5 -X3:3:B
1 mm² BU 1N1-3

1L1-3 / 7.3 -X2:3:B
2.5 mm² BK 1L1-3

-F4
BMELES1705
MCB SINGLE POLE
2A SIEMENS MAKE
5SL61027RC

1.5 mm² BK 2L1-3
-K6:11 23.1 / 2L1-3

1L1-10 / 7.3 -X2:5:A
2.5 mm² BK 1L1-10

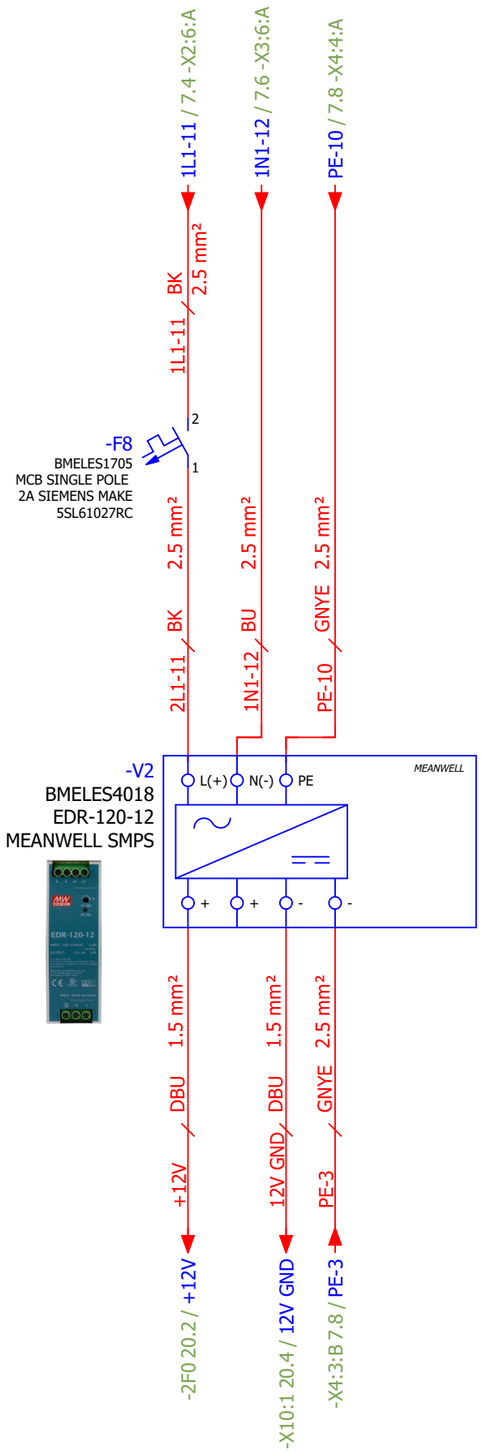
-F7
BMELES1706
MCB 4A ,1 POLE
SIEMENS 5SL61047RC;

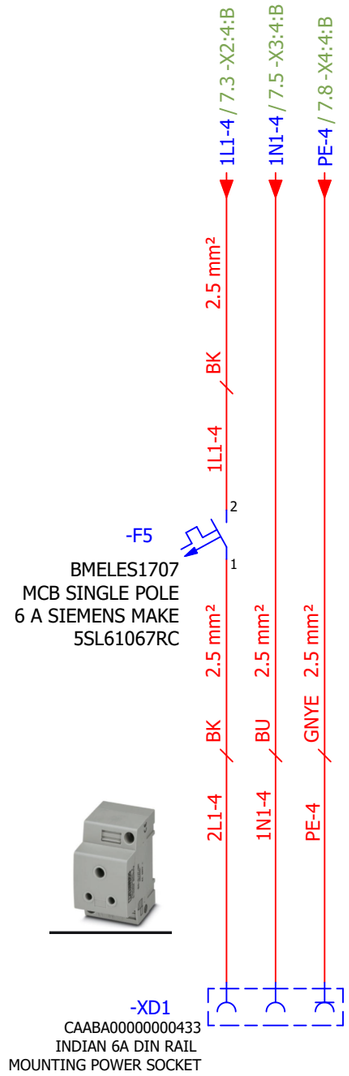
1.5 mm² BK/1 2L1-10
-VM1:6 21.1 / 2L1-10

7 9

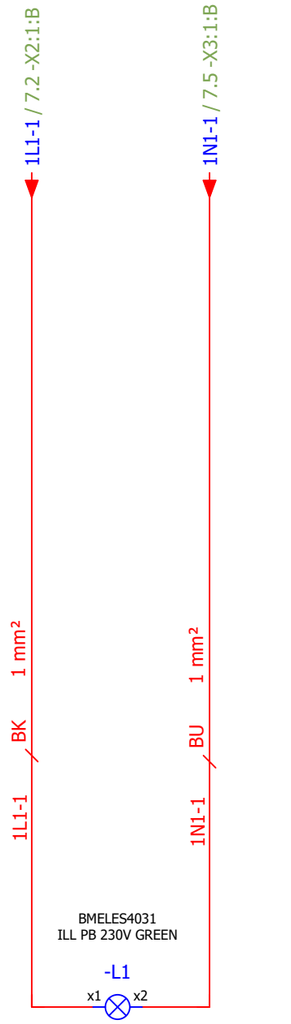
REVISION			Date	10-04-2024	ROBIS	Motherson Automotive & Technology engineering	POWER SUPPLY		=	
			Ed.	MANORAJ.SELVARAJ					+ CP	
			Appr.		S220 HL HV & LV FINAL INSPECTION STATION 11				FMSPM00865	Page 8
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12V SMPS & INDICATION LAMP





POWER SOCKET

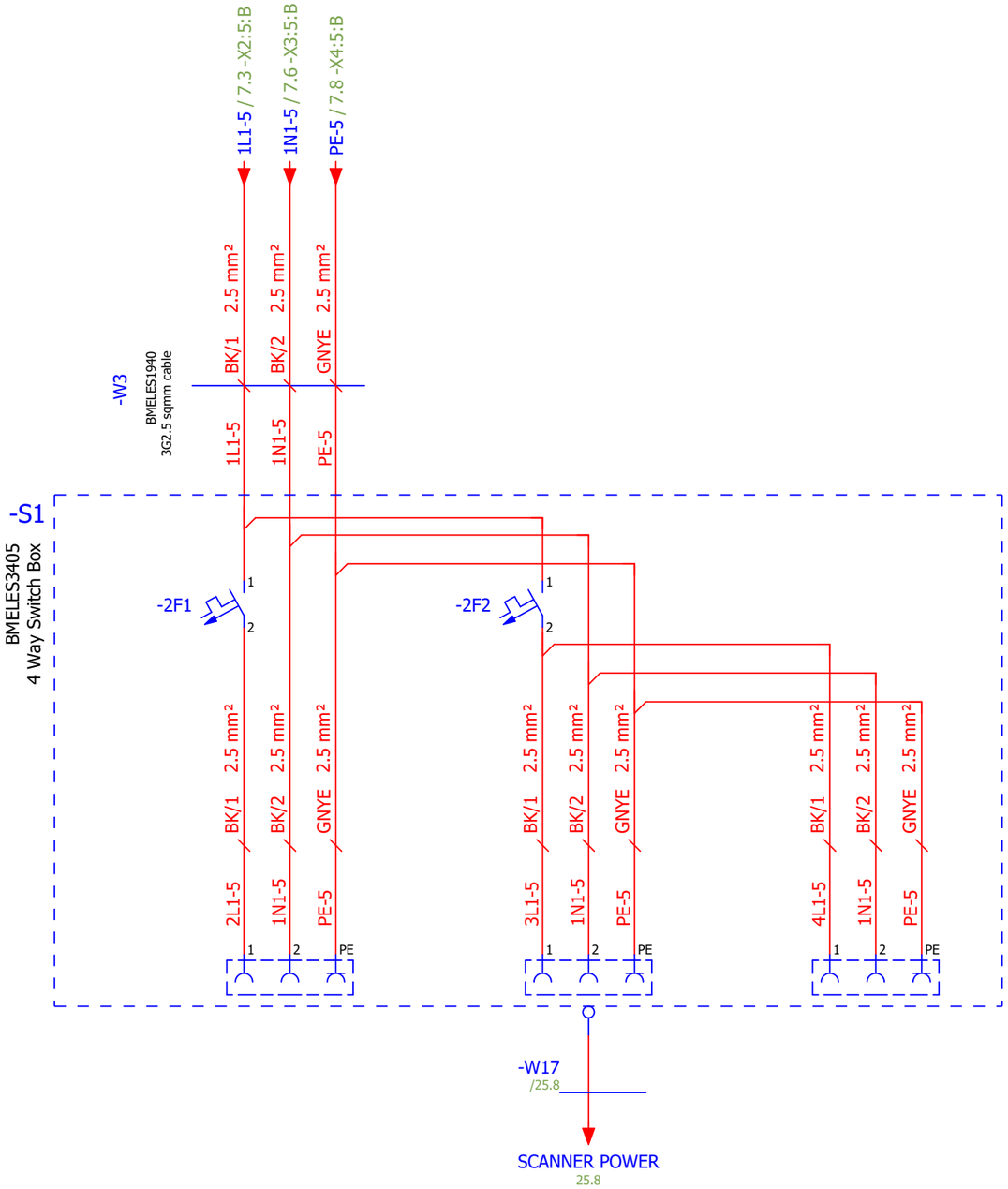


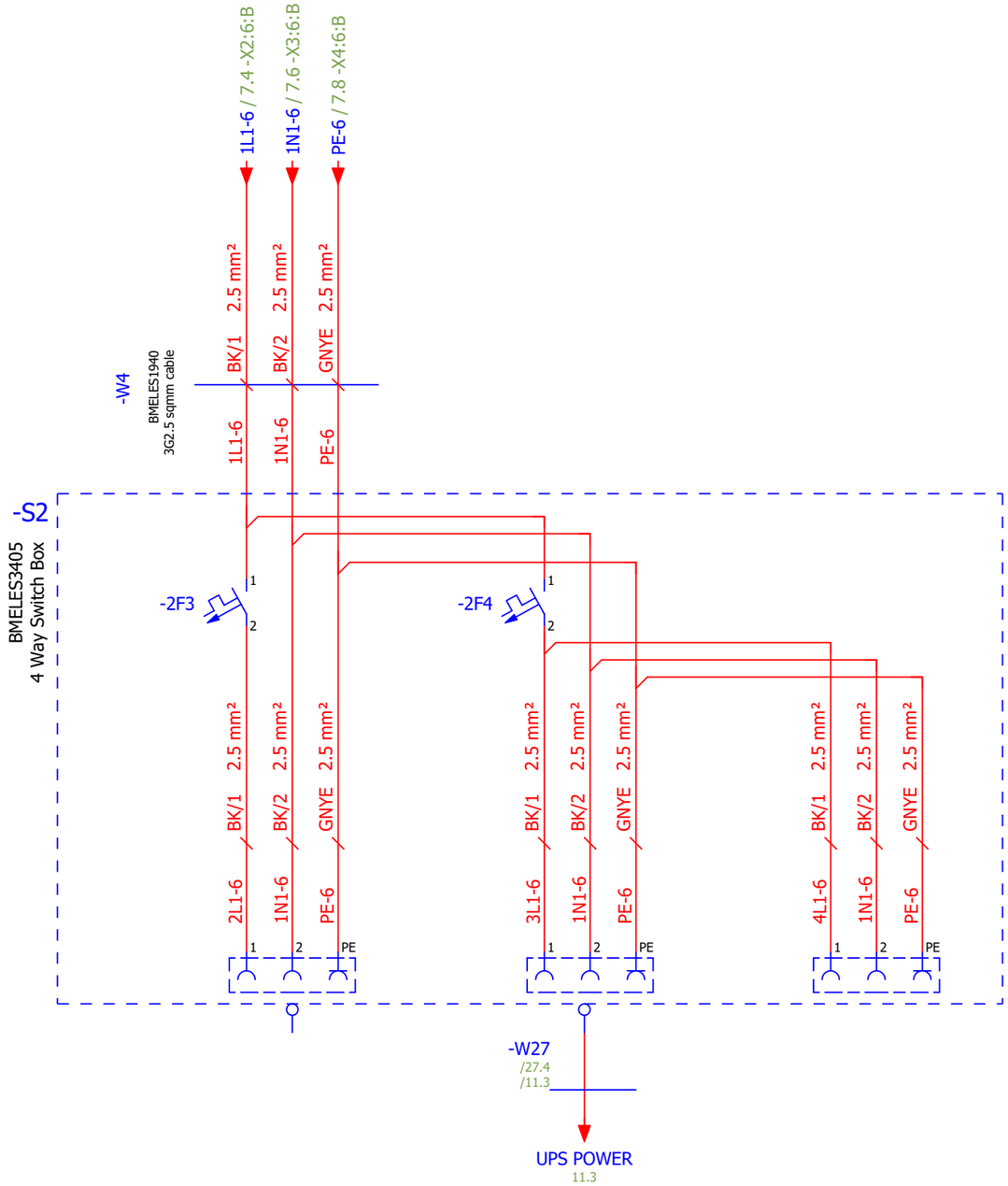
230V GREEN
POWER INDICATION
LAMP

8 10

REVISION			Date	10-04-2024	ROBIS		Motherson Automotive & Technology engineering	12V SMPS & INDICATION LAMP	=
			Ed.	MANORAJ.SELVARAJ					+ CP
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SOCKET INTERFACE





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			Ed.	MANORAJ.SELVARAJ					+ CP	
			Appr.		S220 HL HV & LV FINAL INSPECTION STATION 11			FMSPM00865	Page	10
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UPS INTERFACE



UPS POWER / 10.7 /

-W27

-UPS
BMELES2946
APC 1.1kva Line Interactive
UPS with Built In Battery Backup

L 3 4

L 1 2 | L 1 2 | L 1 2 | L 1 2 | L 1 2

-P1
BMELES3265

-W27.1
BMELES1938
3G1.5 sqmm cable

BK/1 1.5 mm² L01
BK/2 1.5 mm² N01
GNYE 1.5 mm² PE01

-F2:2 8.1 / L01
-V1:N(-) 8.1 / N01
-V1:PE 8.1 / PE01

TO SMPS

-W27.2
POWER CABLE

MONITOR POWER
27.4

TO MONITOR

10 12

REVISION			Date	10-04-2024	ROBIS		Motherson Automotive & Technology engineering	UPS INTERFACE		=
			Ed.	MANORAJ.SELVARAJ						+ CP
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DISTRIBUTION 24VDC TERMINAL BLOCK



-1F0
BMELES1708
MCB SINGLE POLE 10A
SIEMENS MAKE
5SL61107RC

P24VDC1 / 12.2 -X5:1
P24VDC4 / 12.3 -X5:2
24VDC / 8.1 -V1:+
P0VDC / 8.1 -V1:-

P24VDC1 DBU 1.5 mm²
P24VDC4 DBU 1.5 mm²
24VDC DBU 1.5 mm²
P24VDC DBU 1.5 mm²
P0VDC DBU 1.5 mm²

BMELES0584
TB4+HESILED
24 (5X20) EI

-1F1 2A
-1F2 2A

1P24VDC BK/1 0.75 mm²
2P24VDC BK/1 1.5 mm²

-HMI:+ 24.1 / 1P24VDC
26.3 / 2P24VDC

BMELES0364
TERMINAL BLOCK
2.5 SQ MM 1 IN 2 OUT

-X5 1 2 3 4 5

P24VDC1 DBU 1.5 mm² / -1F1 12.0 / P24VDC1
P24VDC2 DBU 0.5 mm² / -1A00:COM0 16.1 / P24VDC2
P24VDC3 DBU 0.5 mm² / -1A00:COM2 16.5 / P24VDC3
P24VDC4 DBU 1.5 mm² / -1F2 12.0 / P24VDC4
P24VDC5 BK/1 0.75 mm² / -PB1-S4:13 28.4 / P24VDC5
P24VDC6 DBU 0.5 mm² / -A1:US:1 13.1 / P24VDC6
P24VDC7 BK/2 0.75 mm² / -PB1-S3:13 28.5 / P24VDC7

-X6 1 2 3 4 5

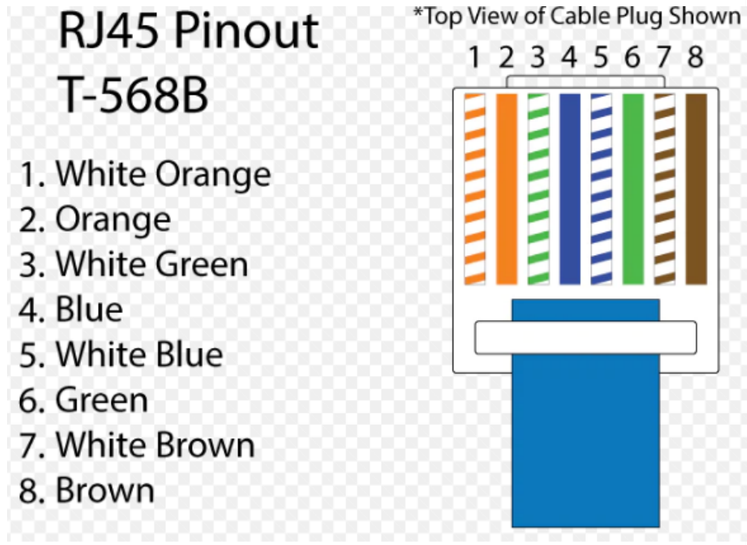
P0VDC1 DBU 0.5 mm² / -1A00:S/S 15.1 / P0VDC1
P0VDC2 BK/2 1.5 mm² / 26.3 / P0VDC2
P0VDC3 DBU 0.5 mm² / -K6:A2 23.1 / P0VDC3
P0VDC4 BK/2 0.75 mm² / -HMI:- 24.2 / P0VDC4
P0VDC5 DBU 0.5 mm² / -K1:A2 18.1 / P0VDC5
P0VDC6 DBU 0.5 mm² / -A1:US:2 13.1 / P0VDC6

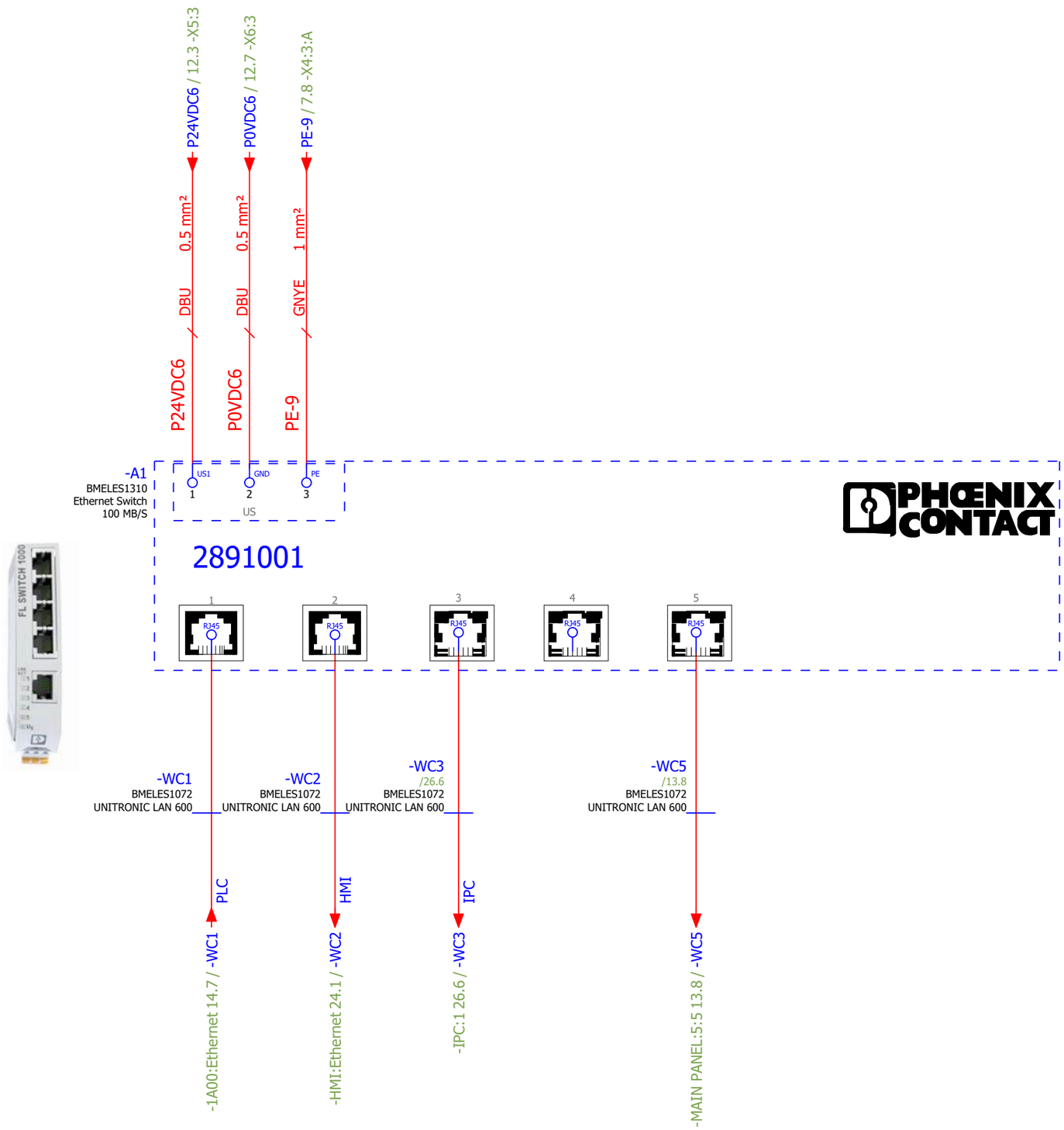
11

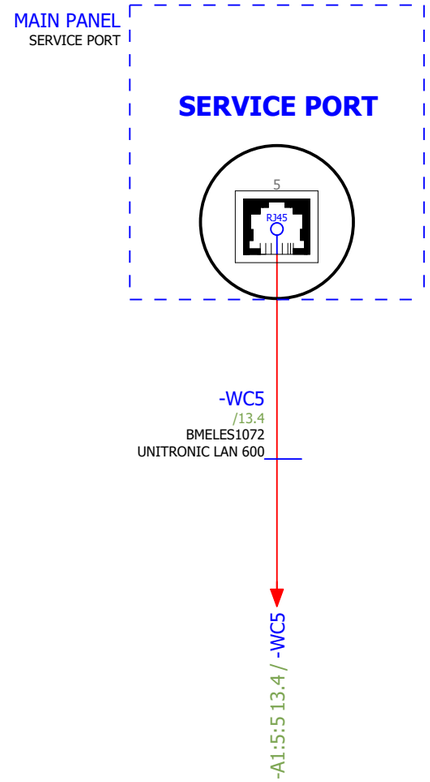
13

REVISION			Date	10-04-2024	ROBIS	Motherson Automotive & Technology engineering	DISTRIBUTION 24VDC TERMINAL BLOCK		=	
			Ed.	MANORAJ.SELVARAJ					+ CP	
			Appr.		S220 HL HV & LV FINAL INSPECTION STATION 11			FMSPM00865	Page	12
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COMMUNICATION







REVISION			Date	10-04-2024	ROBIS	Motherson Automotive & Technology engineering	COMMUNICATION		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV FINAL INSPECTION STATION 11			FMSPM00865	Page 13
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MAIN PLC POWER SUPPLY





1L1-7 / 7.3 -X2:2:A
1N1-7 / 7.6 -X3:7:B
PE-7 / 7.9 -X4:7:B
RS 485+ / 20.8 -X12:1
RS 485- / 20.8 -X12:3
WC1 / 13.1 -A1:1:1

BK 2.5 mm²
1L1-7
2
-F6
1
BMELES1705
MCB SINGLE POLE
2A SIEMENS MAKE
5SL61027RC

BK 2.5 mm² 2L1-7
BU 2.5 mm² 1N1-7
GNYE 2.5 mm² PE-7
BN 0.34 mm² RS 485+
YE 0.34 mm² RS 485-
BN 0.34 mm² RS 485+
YE 0.34 mm² RS 485-

-WC1
/13.1
/24.1
BMELES1072
2170144-UNITRONIC
LAN 600S/FTP CAT7Y

-1A00
CAAAZ00000000077
PLC
FX5U32MRES
MAKE :- MITSUBHISHI

L | N | PE | SG (GND) | SDB (TXD-) | SDA (TXD+) | RDB (RXD-) | RDA (RXD+) | V1+ (PEW0) | V2+ (PEW2) | V-' (PEW4) | V+ (PAW0) | V- (PAW2) | Ethernet

Ethernet port

MITSUBISHI ELECTRIC

Mitsubishi FX5U-32MR-ES 16DI/16DO Part 1 of 5

REVISION			Date	10-04-2024	ROBIS	Motherson Automotive & Technology engineering	MAIN PLC POWER SUPPLY		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV FINAL INSPECTION STATION 11			FMSPM00865	Page 14
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MAIN INPUT MODULE



-1A00
CAAAZ00000000077
PLC FX5U32MRES
MAKE :- MITSUBHISHI

P0VDC1 / 12.6 / -X6:1

P0VDC1 — DBU — 0.5 mm²

S/S | 0V | 24V

X00 / 17.0 -X7:1:A	X01 / 17.0 -X7:1:C	X02 / 17.0 -X7:2:A	X03 / 17.1 -X7:2:C	X04 / 17.1 -X7:3:A	X05 / 17.1 -X7:3:C	X06 / 17.1 -X7:4:A	X07 / 17.1 -X7:4:C	X10 / 17.1 -X7:5:A	X11 / 17.1 -X7:5:C	X12 / 17.2 -X7:6:A	X13 / 17.2 -X7:6:C	X14 / 17.2 -X7:7:A	X15 / 17.2 -X7:7:C	X16 / 17.2 -X7:8:A	X17 / 17.2 -X7:8:C
DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU
0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²
X00	X01	X02	X03	X04	X05	X06	X07	X10	X11	X12	X13	X14	X15	X16	X17
X00	X01	X02	X03	X04	X05	X06	X07	X10	X11	X12	X13	X14	X15	X16	X17
E0.0	E0.1	E0.2	E0.3	E0.4	E0.5	E0.6	E0.7	E1.0	E1.1	E1.2	E1.3	E1.4	E1.5	E1.6	E1.7
EMERGENCY STOP PB	RH CYCLE START PB	SCREW DRIVER OK ALL	AMMETER	FIXTURE SPARE	FIXTURE SPARE	LB
LH CYCLE START PB	RESET	SPARE	SCREW DRIVER NG	TURN	FIXTURE SPARE	FIXTURE SPARE	HB.

MITSUBISHI ELECTRIC
Mitsubishi FX5U-32MR-ES 16DI/16DO Part 2 of 5

-1A00

X10	X11	X12	X13	X14	X15	X16	X17		
E1.0	E1.1	E1.2	E1.3	E1.4	E1.5	E1.6	E1.7		
AMMETER	FIXTURE SPARE	FIXTURE SPARE	LB						
TURN	FIXTURE SPARE	FIXTURE SPARE	HB.						

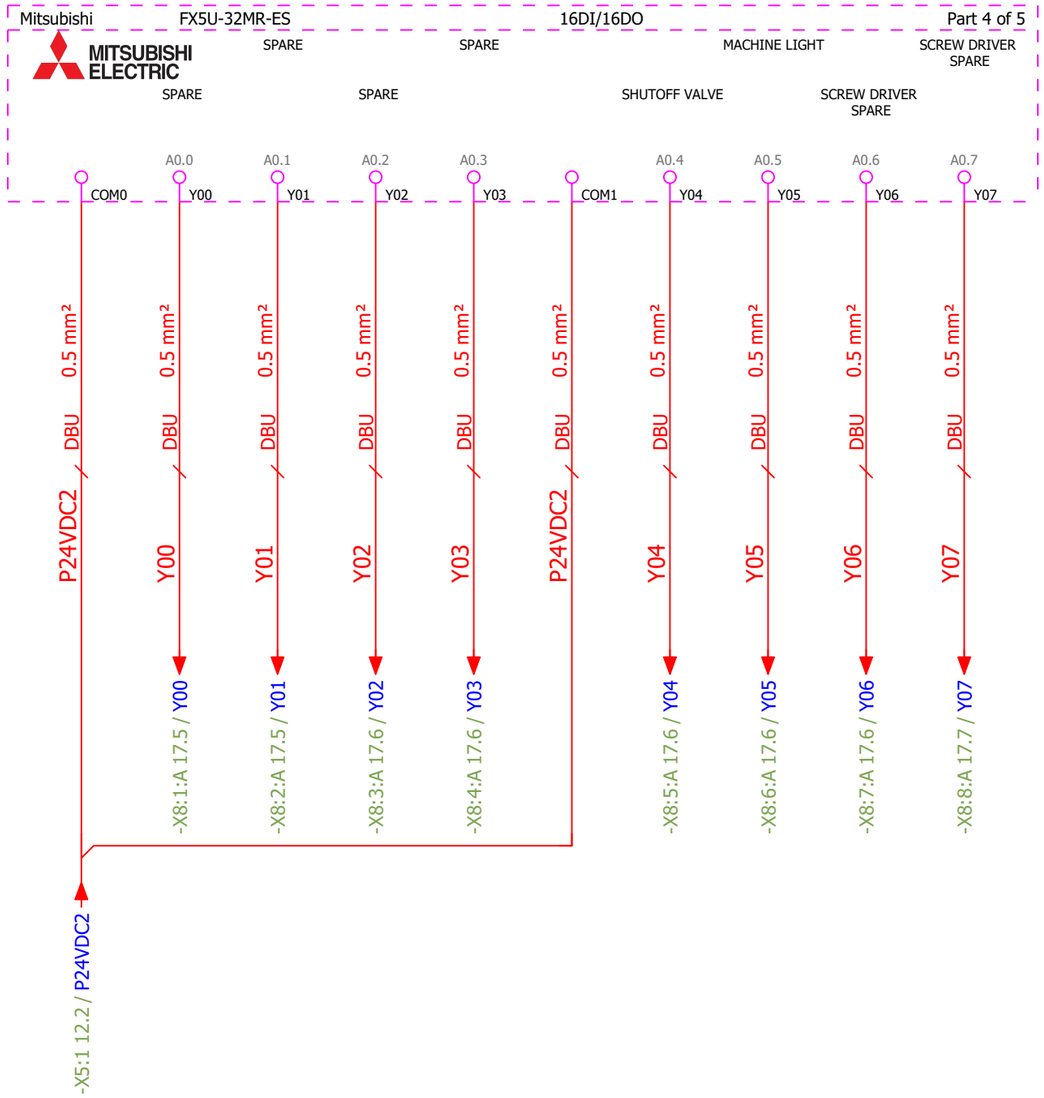
MITSUBISHI ELECTRIC
Mitsubishi FX5U-32MR-ES 16DI/16DO Part 3 of 5

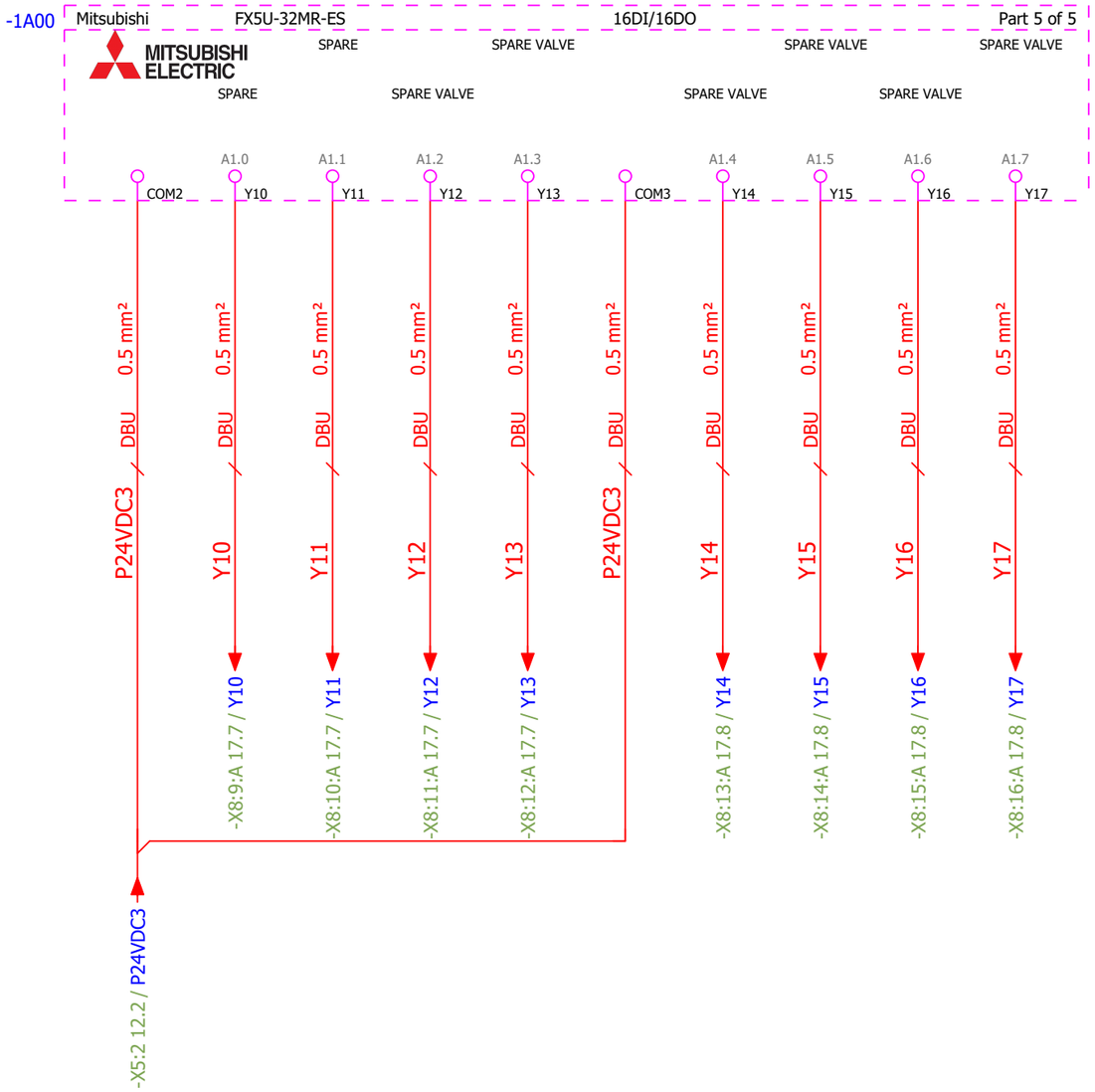
REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	MAIN INPUT MODULE		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV FINAL INSPECTION STATION 11			FMSPM00865	Page 15
Modification	Date	Name	Original		Replacement of	Replaced by			Page 15 / 29

MAIN OUTPUT MODULE



-1A00
CAAAZ00000000077
PLC FX5U32MRES
MAKE :- MITSUBHISHI

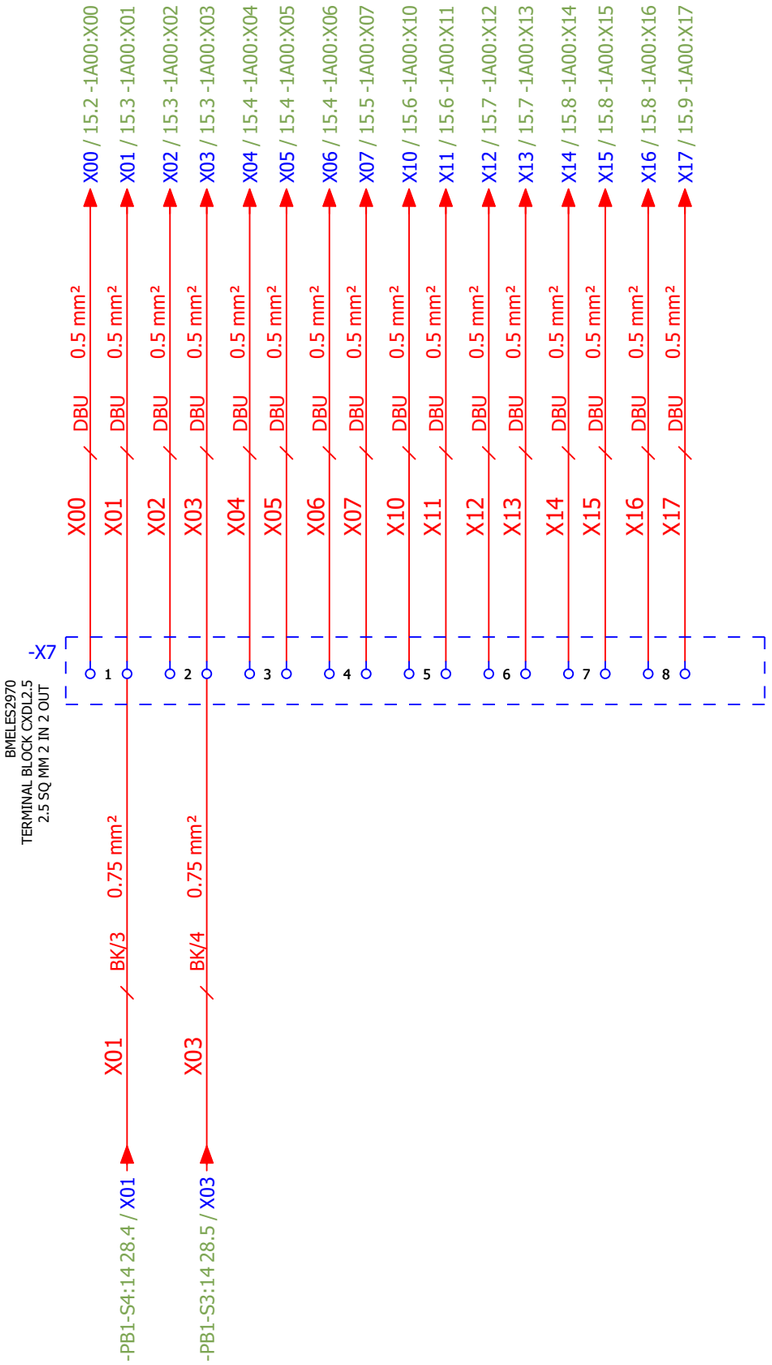


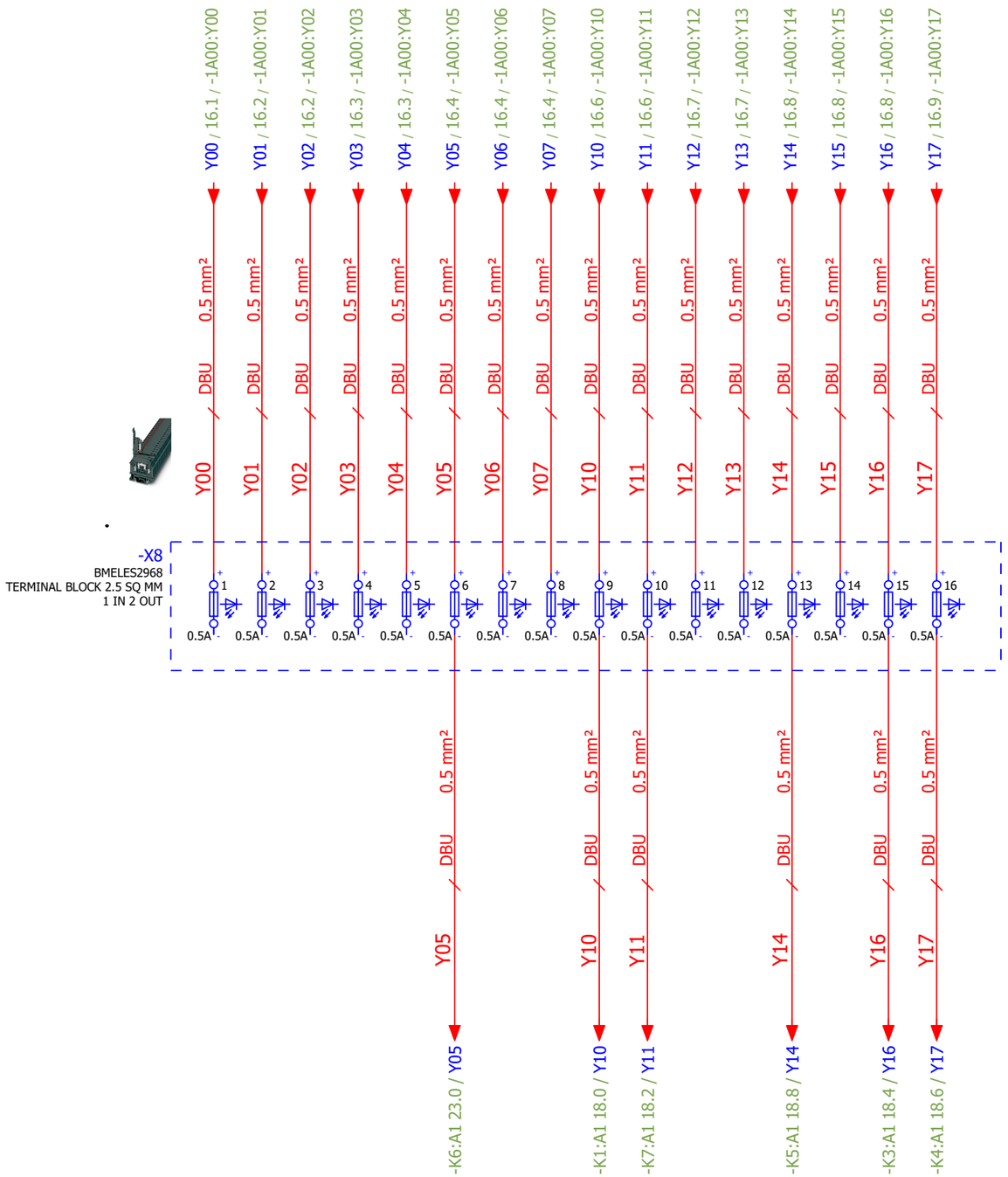


REVISION			Date	10-04-2024	ROBIS	Motherson Automotive & Technology engineering	MAIN OUTPUT MODULE	=
			Ed.	MANORAJ.SELVARAJ				+ CP
			Appr.		S220 HL HV & LV FINAL INSPECTION STATION 11		FMSPM00865	Page 16
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INPUT & OUTPUT TERMINAL DISTRIBUTION

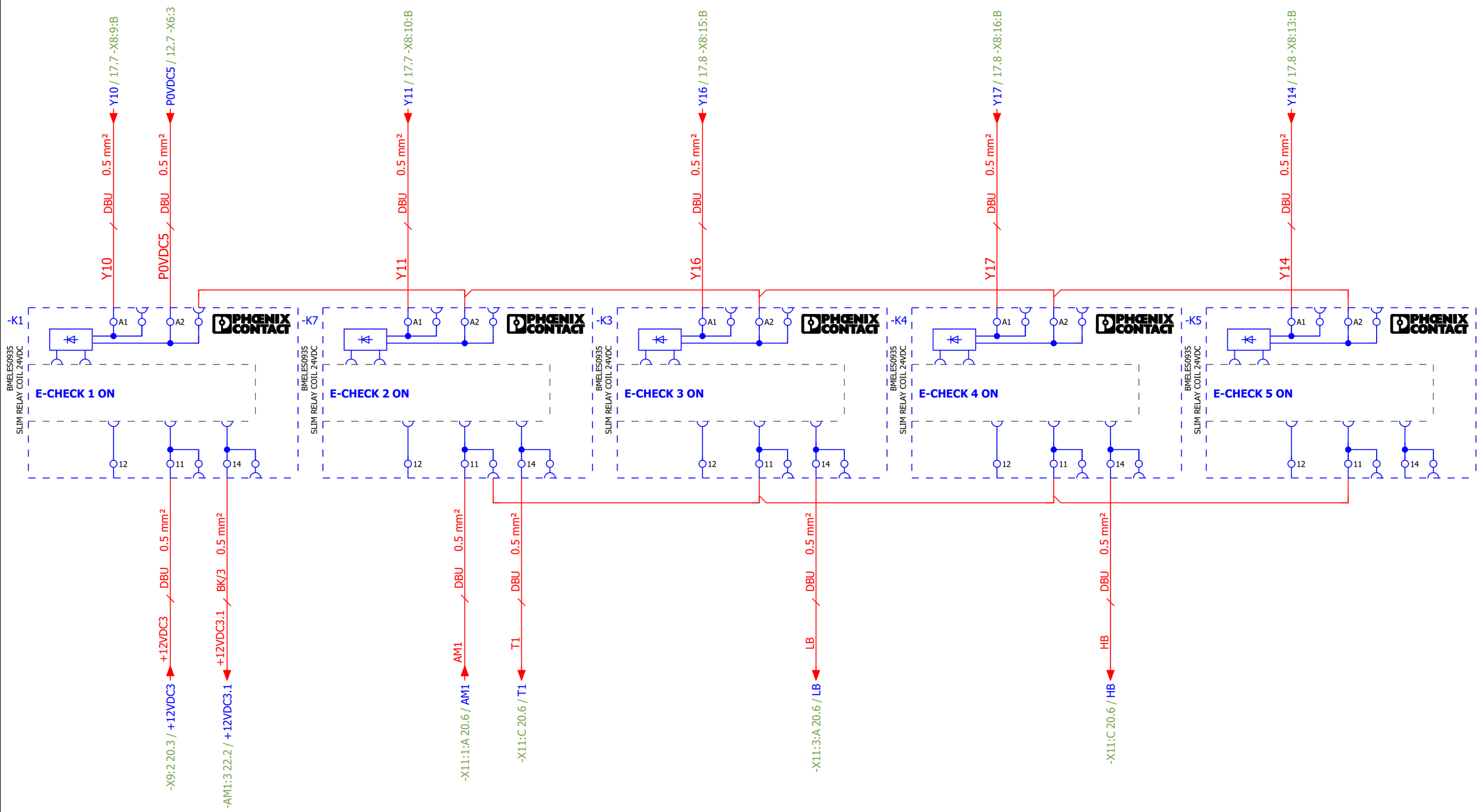
ALL FUSE SHOULD BE 0.5 A





REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	INPUT & OUTPUT TERMINAL DISTRIBUTION		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV FINAL INSPECTION STATION 11			FMSPM00865	Page 17
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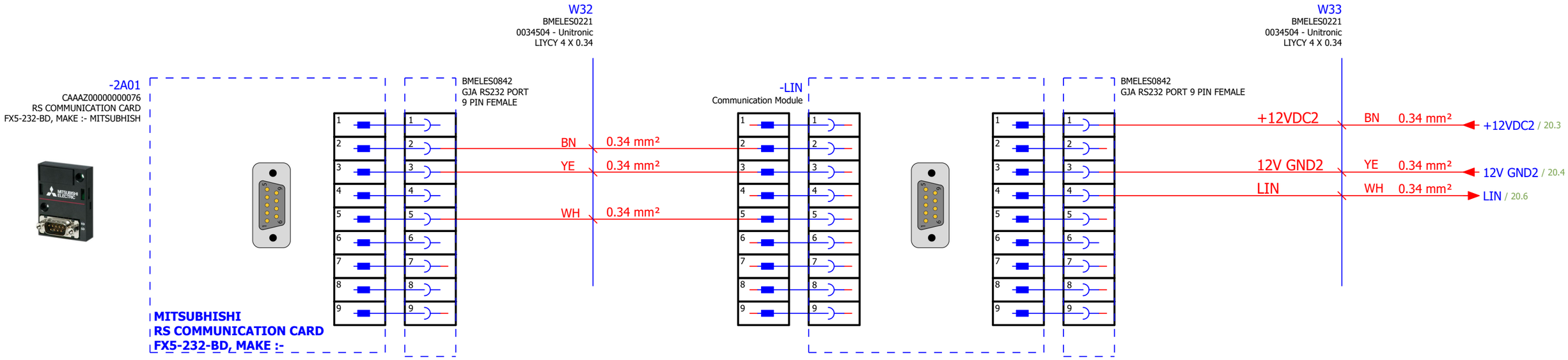
AMMETER RELAY INTERFACE



17 19

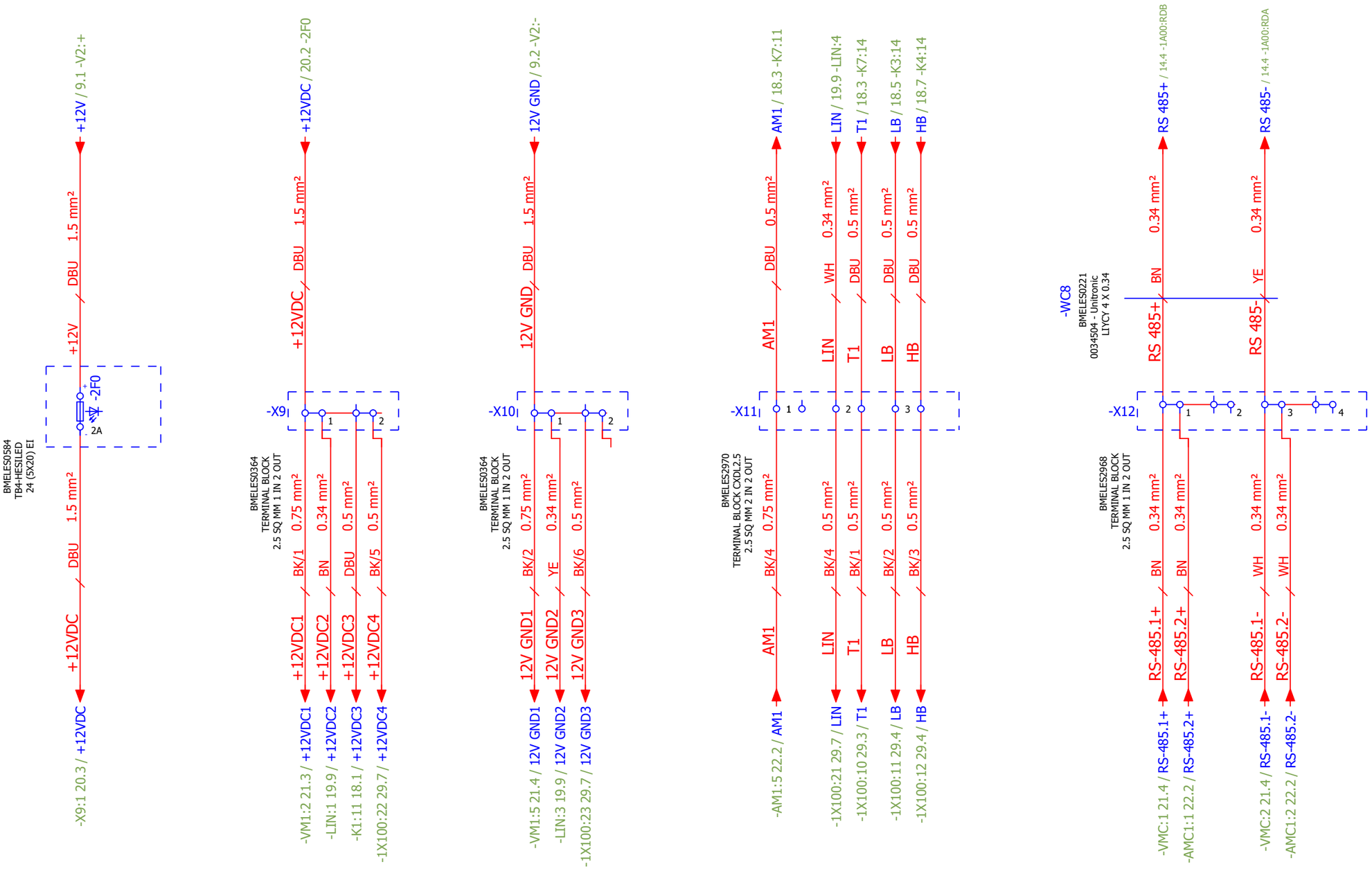
REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	AMMETER RELAY INTERFACE		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV FINAL INSPECTION STATION 11			FMSPM00865	Page 18
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LIN COMMUNICATION MODULE INTERFACE



REVISION			Date	10-04-2024	ROBIS	Motherson Automotive & Technology engineering	LIN COMMUNICATION MODULE INTERFACE		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV FINAL INSPECTION STATION 11			FMSPM00865	Page 19
Modification	Date	Name	Original		Replacement of	Replaced by			Page 19 / 29

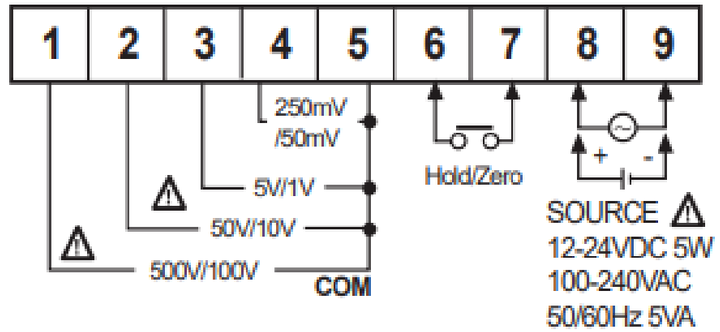
12VDC DISTRIBUTION & AMMETER COM

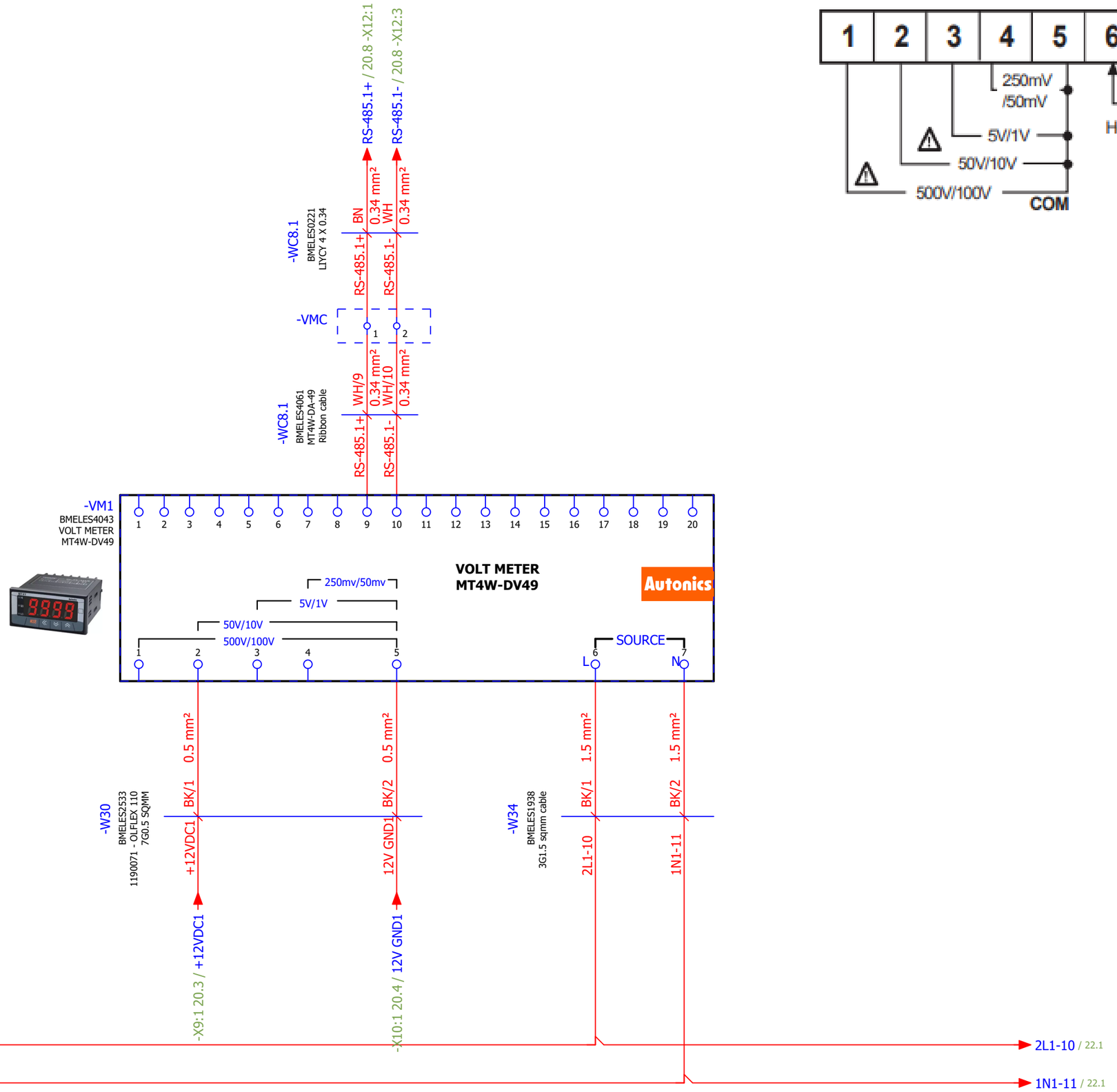


REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	12VDC DISTRIBUTION & AMMETER COM		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV FINAL INSPECTION STATION 11			FMSPM00865	Page 20
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DIGITAL PANEL METER

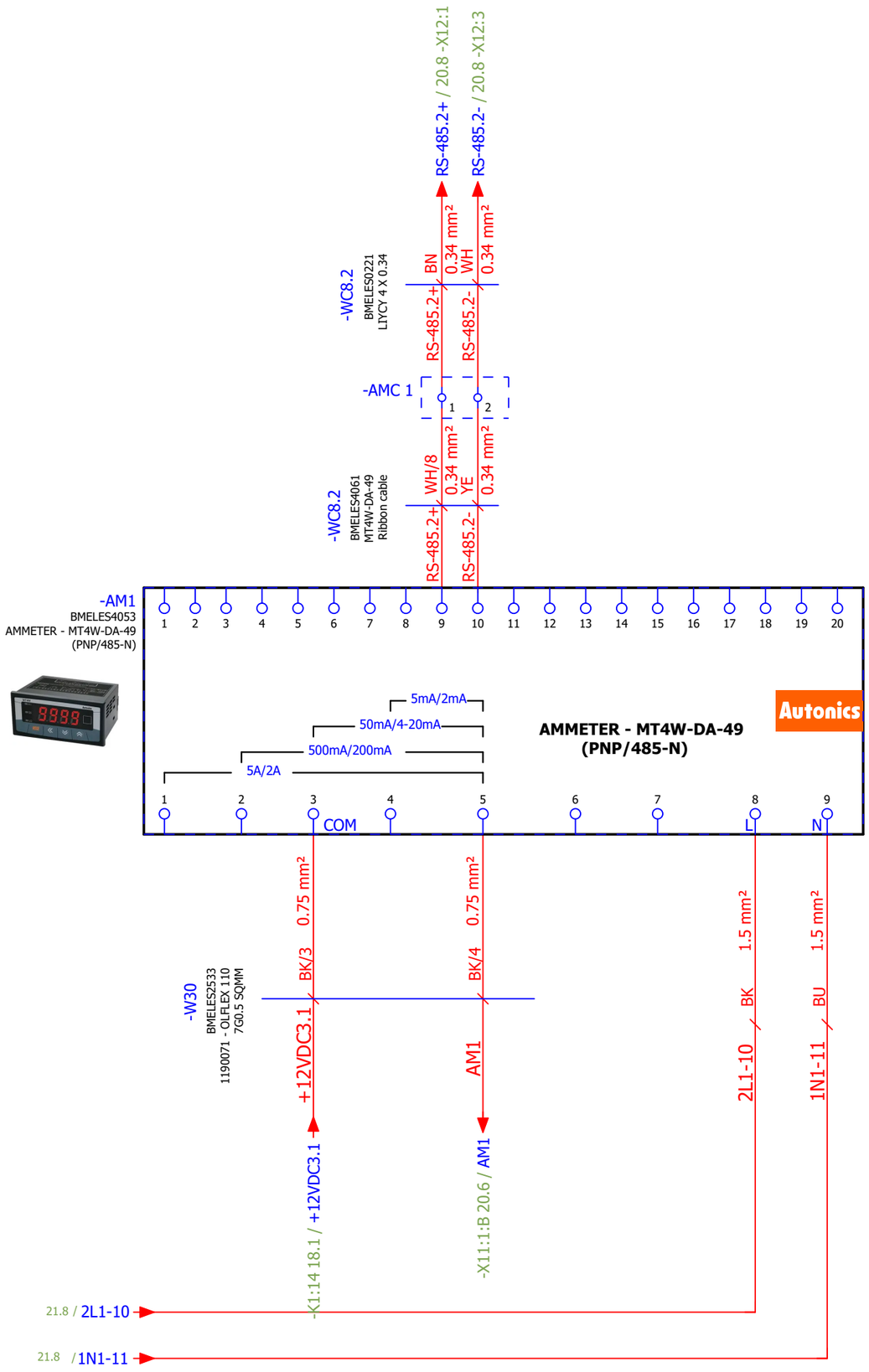
• MT4W-DV- □□



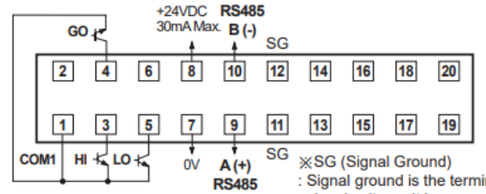


REVISION			Date	10-04-2024	ROBIS		Motherson Automotive & Technology engineering	DIGITAL PANEL METER	=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV FINAL INSPECTION STATION 11			FMSPM00865	Page 21
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AMMETER INTERFACE



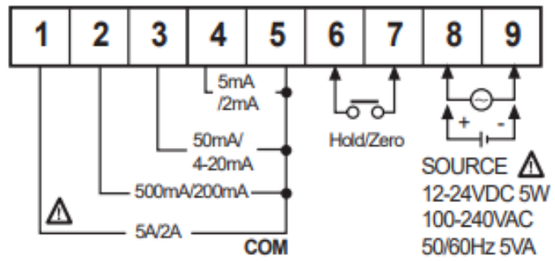
MAIN OUT:
NPN OPEN COLLECTOR
12-24VDC Max. 50mA



MAIN OUT:
PNP OPEN COLLECTOR
12-24VDC Max. 50mA

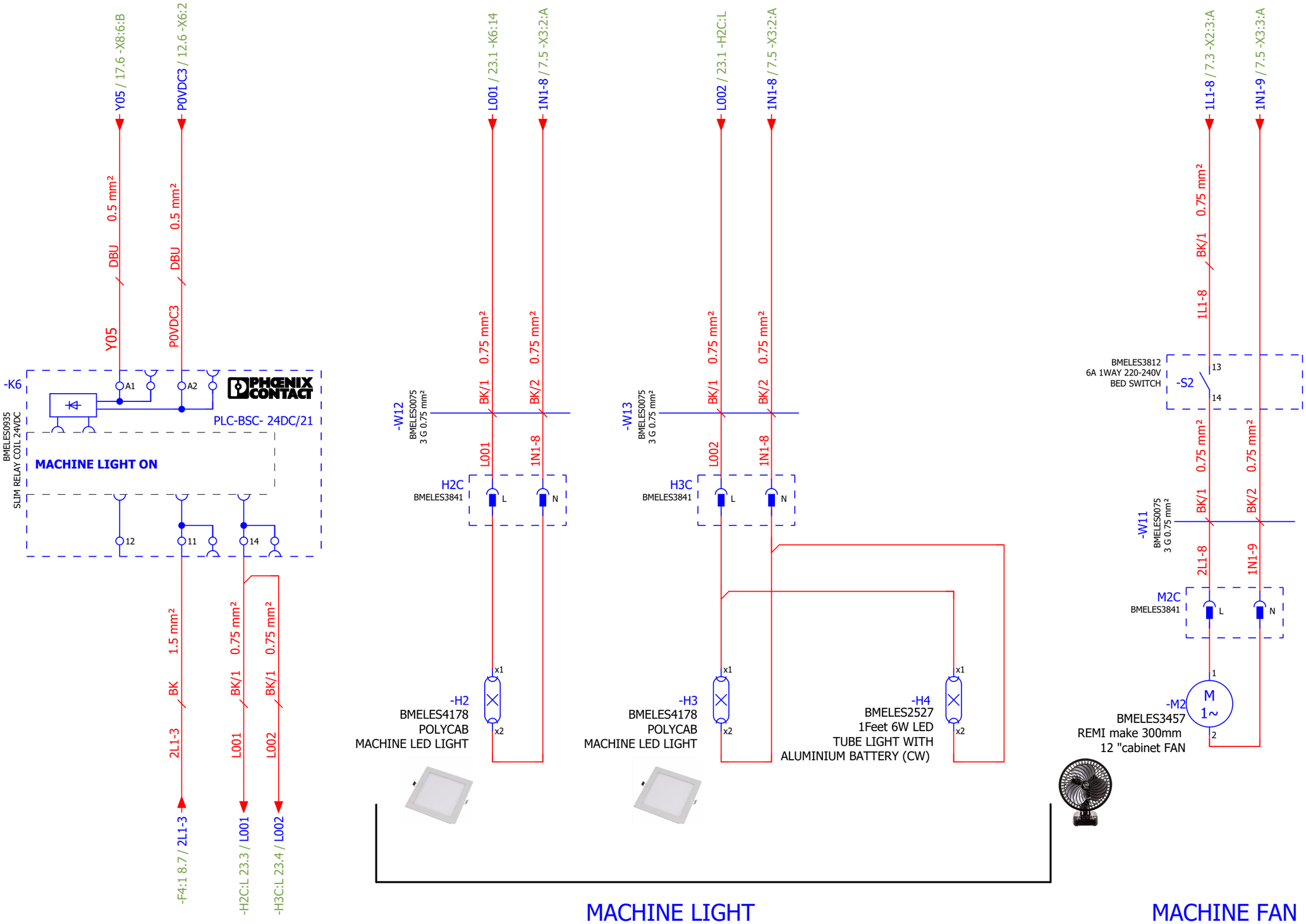
※SG (Signal Ground)
: Signal ground is the terminal which is base of the signal voltage. It is connected with terminal 7 inside of the product.

- **MT4W-DA-□□**



REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	AMMETER INTERFACE		=
			Ed.	MANORAJ.SELVARAJ	S220 HL HV & LV FINAL INSPECTION STATION 11				+ CP
			Appr.					FMSPM00865	Page 22
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MACHINE FAN INTERFACE



MACHINE LIGHT

MACHINE FAN

REVISION			Date	10-04-2024	ROBIS	Motherson Automotive & Technology engineering	MACHINE FAN INTERFACE		=
			Ed.	MANORAJ.SELVARAJ	S220 HL HV & LV FINAL INSPECTION STATION 11				+ CP
			Appr.					FMSPM00865	Page 23
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HMI INTERFACE



-WC2 / 13.2 -A1:2:2

-WC6 / 25.2 -HMI RS232-COM1

-WC1
BMELES1072
UNITRONIC LAN 600

-WC6
BMELES0221
0034504 - Unitronic
LIYCY 4 X 0.34

-HMI
CAAAI00000000586
HMI GS2107-WTBD

Ethernet | RS-232 | RS-422 | USB

MITSUBISHI ELECTRIC

+ | - | FG

-W5
BMELES1937
3 G 0.75 mm²

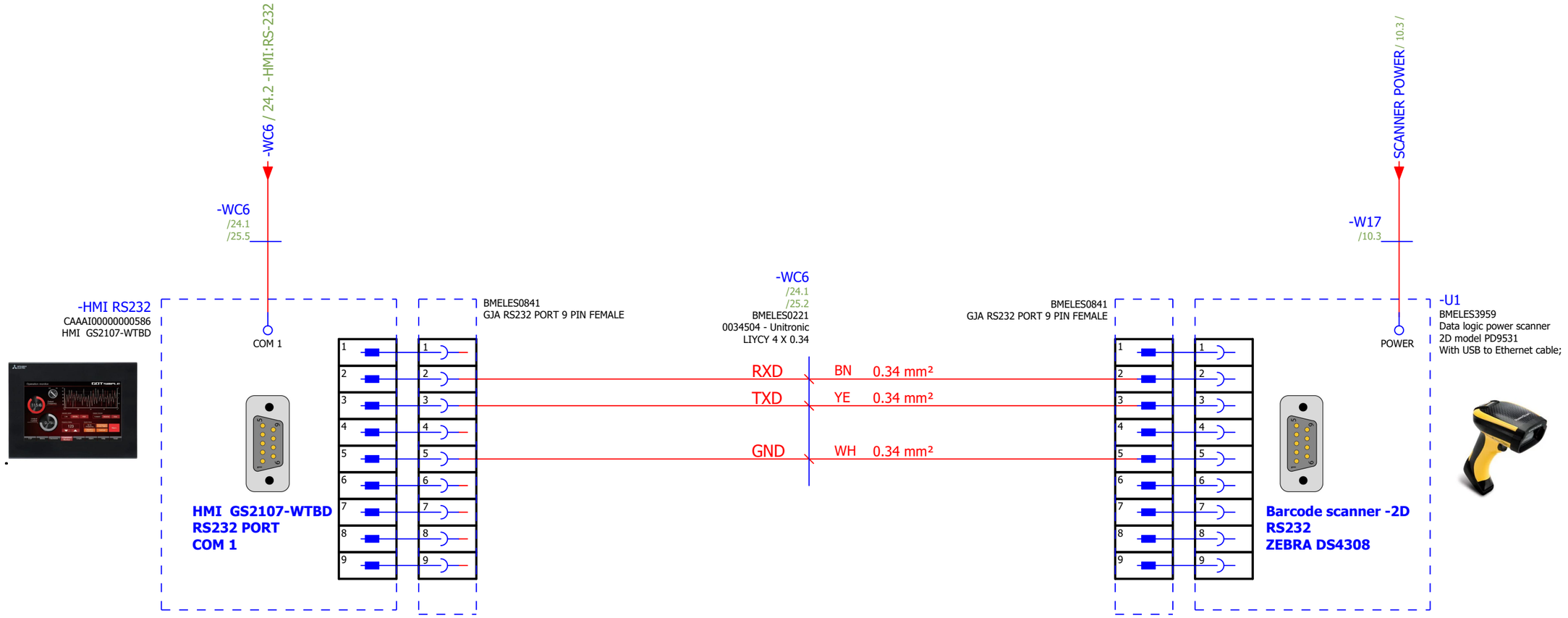
	+	-	FG
	0.75 mm²	0.75 mm²	0.75 mm²
	BK/1	BK/2	GNYE
	1P24VDC	P0VDC4	PE-8
	-1F1 / 12.0 / 1P24VDC	-X6:2 / 12.6 / P0VDC4	-X4:2:A / 7.7 / PE-8

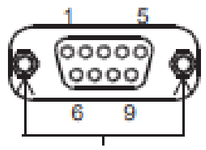
HMI POWER SUPPLY

REVISION			Date	10-04-2024	ROBIS	Motherson Automotive & Technology engineering	HMI INTERFACE		=	
			Ed.	MANORAJ.SELVARAJ					+ CP	
			Appr.		S220 HL HV & LV FINAL INSPECTION STATION 11				FMSPM00865	Page 24
Modification	Date	Name	Original		Replacement of	Replaced by				Page 24 / 29

0	1	2	3	4	5	6	7	8	9

HMI TO SCANNER COMMUNICATION





Screws to fix
RS-232C connector
#4-40UNC
(inch screw thread)

Pin No.	Signal	Name
1	CD (DCD)	Receive carrier detection
2	RD (RXD)	Receive data
3	SD (TXD)	Send data
4	ER (DTR)	Send request
5	SG (GND)	Signal ground
6	DR (DSR)	Send enabled
7, 8, 9		Not used

REVISION			Date	10-04-2024	ROBIS	Motherson Automotive & Technology engineering	HMI TO SCANNER COMMUNICATION		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV FINAL INSPECTION STATION 11			FMSPM00865	Page 25
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IPC INTERFACE

HDMI / 27.6 -MONITOR-HDMI

-WC9
HDMI CABLE

WC3 / 13.2 -A1:3:3

-WC3
BMELES1072
UNITRONIC LAN 600

-IPC

BMCONS0728
MOOTEK Industrial PC
MBOX-520-I5-M7-8265U



DIGITAL I/O





8- CHL
DIGITAL I/O

HDMI

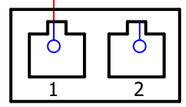




COM 1









-USB PORT1







P1
POWER

VGA

COM 2

-W35
BMELES1938
3G1.5 sqmm cable

1.5 mm²
BK/1
2P24VDC
-1F2 12.0 / 2P24VDC

1.5 mm²
BK/2
P0VDC2
-X6:1 12.6 / P0VDC2

-WC10
USB A to MINI USB CABLE

-MONITOR-TOUCH / 27.4 / TOUCH

REVISION			Date	10-04-2024	ROBIS	Motherson Automotive & Technology engineering	IPC INTERFACE		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV FINAL INSPECTION STATION 11			FMSPM00865	Page 26
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MONITOR INTERFACE

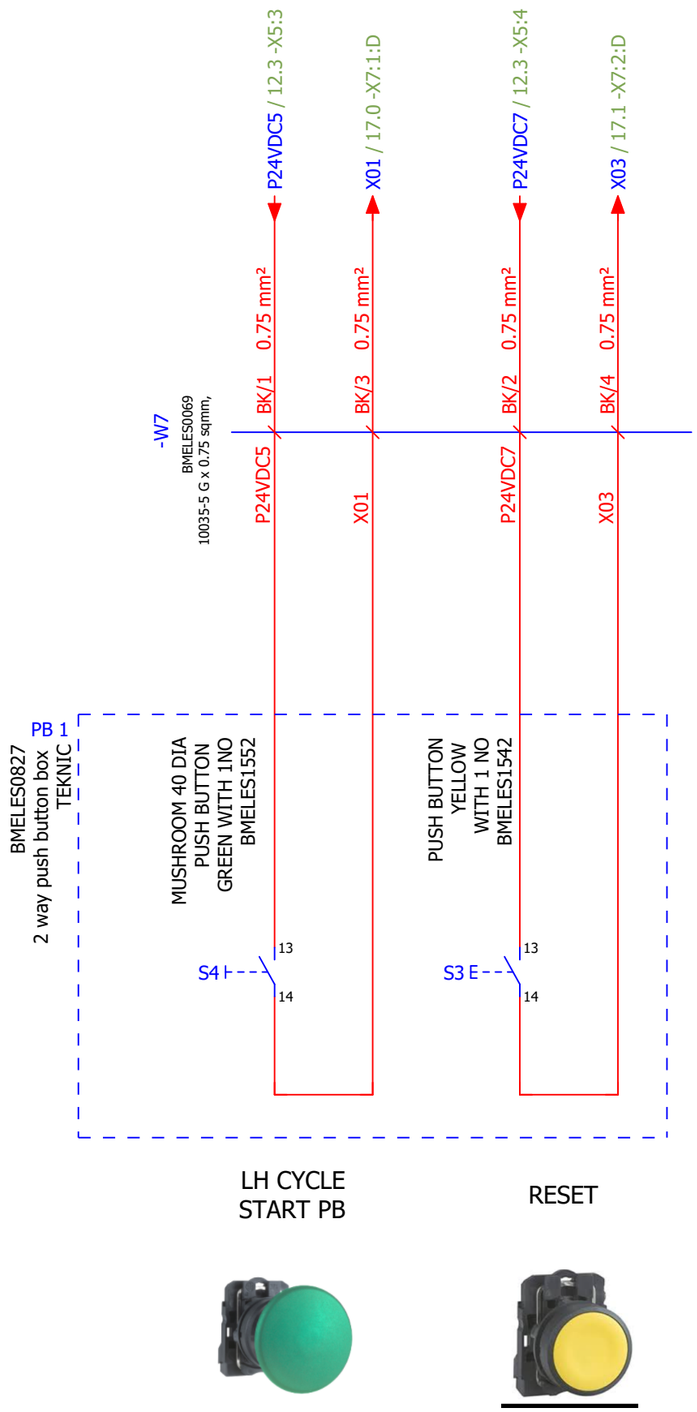
-MONITOR
BMELES3308
22"Touch Monitor





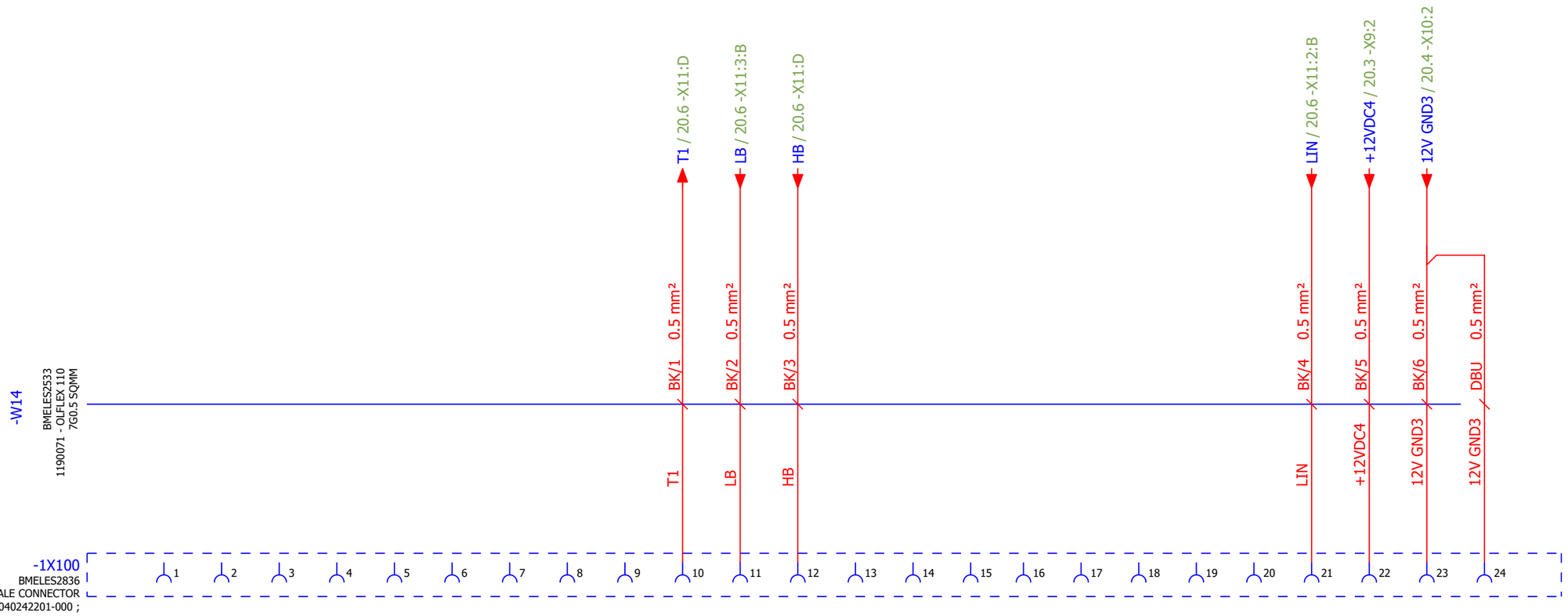
REVISION			Date	10-04-2024	ROBIS	Motherson Automotive & Technology engineering	MONITOR INTERFACE		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV FINAL INSPECTION STATION 11			FMSPM00865	Page 27
Modification	Date	Name	Original		Replacement of	Replaced by			Page 27 / 29

PUSH BUTTON INTERFACE



REVISION			Date	10-04-2024	ROBIS	Motherson Automotive & Technology engineering	PUSH BUTTON INTERFACE		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV FINAL INSPECTION STATION 11			FMSPM00865	Page 28
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24 PIN I/O INTERFACE



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REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	24 PIN I/O INTERFACE		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV FINAL INSPECTION STATION 11			FMSPM00865	Page 29
Modification	Date	Name	Original		Replacement of	Replaced by			Page 29 / 29